🦴 BIOLOGY CH.15 — SKELETON SYSTEM — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📑 Topics in this chapter:

1. Movement & Coordination

- Animal movement Muscular + Skeletal + Nervous systems ki coordinated activity hai. Muscles contract, bones lever ki tarah kaam karte hain, nerves signals bhejte hain. *[Q821]*

2. Muscle Cramps & Anaerobic Respiration

- 400m race ke baad cramps isliye hote hain kyunki intense exercise mein oxygen ki kami ho jati hai → anaerobic respiration → lactic acid accumulate hota hai muscles mein → pain/cramps. *[Q822]*
- Aerobic respiration: glucose + oxygen → CO_2 + water + energy. Anaerobic: glucose → lactic acid + energy (kam). *[Q822]*

3. Human Skeleton - Bones & Structure

- Adult human skeleton mein 206 bones hain. Baby mein ~270 bones hote hain, kuch fuse ho jati hain. *[Q823, Q824, Q825]*
- Human skeleton 2 divisions: Axial skeleton (skull, vertebral column, ribs, sternum - 80 bones) aur Appendicular skeleton (limbs, girdles - 126 bones). *[Q826]*
- Skull 22 bones se bana hai. *[Q827]*
- Vertebral column 26 bones (33 baby mein): 7 cervical, 12 thoracic, 5 lumbar, 1 sacrum, 1 coccyx. *[Q828]*
- Rib cage: 12 pairs ribs (24) + sternum (1) = 25 bones. *[Q829]*

4. Types of Bones

- Long bones: limbs mein (humerus, femur) - leverage aur support. *[Q830]*
- Short bones: wrist (carpals), ankle (tarsals) - stability aur limited movement. *[Q830]*

- Flat bones: skull, ribs, sternum - protection aur muscle attachment. *[Q830]*
- Irregular bones: vertebrae, facial bones - complex shapes. *[Q830]*
- Sesamoid bones: tendons ke andar (patella/kneecap) - friction protect karte hain. *[Q830]*

5. Joints & Movements

- Fixed joints (Immovable): skull bones (sutures). *[Q831]*
- Slightly movable: vertebral column, pubic symphysis. *[Q831]*
- Freely movable (Synovial): knee, elbow, shoulder, hip. *[Q831]*
- Ball and socket joint: shoulder, hip - maximum movement. *[Q832]*
- Hinge joint: knee, elbow - bending/straightening. *[Q832]*
- Pivot joint: neck (atlas-axis) - rotation. *[Q832]*
- Gliding joint: wrist, ankle - sliding movement. *[Q832]*
- Saddle joint: thumb - 2 planes mein movement. *[Q832]*

6. Bone Composition & Growth

- Bone composition: Calcium phosphate ($Ca_3(PO_4)_2$) main mineral, collagen fibres, osteocytes (bone cells). *[Q833, Q834]*
- Calcium aur Vitamin D bones ke liye essential hain. *[Q833]*
- Bone growth: epiphyseal plates (growth plates) se hoti hai - long bones ke ends par. *[Q835]*
- Osteoblasts: new bone form karte hain. Osteoclasts: old bone dissolve karte hain. Osteocytes: mature bone cells. *[Q836]*

7. Axial & Appendicular Skeleton

- Axial skeleton: central axis form karta hai - skull, hyoid, vertebral column, thoracic cage. *[Q837]*
- Appendicular skeleton: limbs aur girdles - pectoral girdle (shoulder), pelvic girdle (hip), upper limbs, lower limbs. *[Q837]*

8. Muscular System

- 3 types of muscles: Skeletal (striated, voluntary), Smooth (non-striated, involuntary), Cardiac (striated, involuntary). *[Q838]*
- Skeletal muscles: bones se attached, body movement. ~600 muscles human body mein. *[Q838]*
- Smooth muscles: internal organs (stomach, intestine, blood vessels) - involuntary. *[Q838]*

- Cardiac muscle: heart - involuntary, striated, intercalated discs. *[Q838]*
- Tendons: muscles ko bones se connect karte hain (dense connective tissue). *[Q839]*
- Ligaments: bones ko bones se connect karte hain joints par (elastic connective tissue). *[Q839]*

9. Bone Disorders & Diseases

- Osteoporosis: bones weak aur brittle ho jati hain - calcium loss. Common in elderly/post-menopausal women. *[Q840]*
- Arthritis: joints ka inflammation - pain, swelling, stiffness. Types: Osteoarthritis, Rheumatoid arthritis, Gout. *[Q841]*
- Rickets: children mein Vitamin D/calcium deficiency se bones soft aur weak ho jati hain. *[Q842]*
- Osteomalacia: adults mein Vitamin D deficiency se bone softening. *[Q842]*
- Fracture: bone break - simple, compound, comminuted, greenstick. *[Q843]*

10. Other Skeletal Facts

- Bone marrow: red marrow (blood cells produce karta hai) aur yellow marrow (fat store). *[Q844]*
- Longest bone: Femur (thigh bone). Smallest bone: Stapes (ear mein). *[Q845]*
- Strongest bone: Femur. *[Q845]*
- Patella (kneecap): sesamoid bone, tendon mein located. *[Q846]*
- X-rays bones ki imaging ke liye use hoti hain - dense structures (bones) white dikhte hain. *[Q847]*
- cartilage: bones ke ends par, smooth movement aur shock absorption. *[Q848]*

11. Additional Facts — Joints, Muscles, Connective Tissue & Bones

- Middle ear ki 3 bones (ossicles): Hammer (Malleus), Anvil (Incus), aur Stirrup (Stapes) — sound wave pressure variations amplify karke inner ear tak transmit karti hain. *[Q849]*
- 'Synovial fluid' freely movable joints (synovial joints) mein milti hai — shoulders, hips, knees, elbows mein hoti hai, joint capsule hoti hai. Fibrous joints strong collagen tissue se, Cartilaginous joints cartilage se judte hain. *[Q850]*
- 'Hedgehog' apne aap ko protect karne ke liye spines ka coat rakhta hai — spines hedgehogs, tenrecs, echidnas aur rodents mein milte hain. *[Q851]*
- 'Rheumatism' body ke 'joints' ko affect karta hai — autoimmune-inflammatory disease hai, immune system joints lining cells ko attack karta hai, swelling-stiffness-pain hoti hai. *[Q852]*
- Knee cap ko 'Patella' kehte hain — hind limb (leg) ki bones mein Femur (longest bone), Tibia, Fibula, Patella, Tarsals, Metatarsals, Phalanges shaamil hain. *[Q853]*
- Human body ki sabse chhoti muscle 'Stapedius' hai — middle ear mein stapes bone se attached hoti hai. Pectoralis Major chest ki largest muscle, Tibialis leg ki largest muscle hai. *[Q854]*

- Biceps muscle 'Arm' mein hoti hai — shoulder aur elbow ke beech front upper arm mein large muscle hai. Cardiac muscle sirf heart mein, Visceral muscle intestines-blood vessels mein. *[Q855]*
- 'Periodontics' dentistry ki branch hai — gum diseases (jo teeth ko support karne wali structures destroy karti hain) treat karti hai. Gastroenterology stomach-intestine diseases, Orthopedics musculoskeletal system deal karta hai. *[Q856]*
- 2 bones ko connect karne wala connective tissue 'Ligament' kehlaata hai — bone-to-bone connect karta hai, collagen-rich strong fibrous tissue hai. *[Q857]*
- Bone cells 'Calcium (Ca) aur Phosphorus (P)' se bane hard matrix mein embedded hote hain — cortical/compact bone (hard, femur-tibia jaisi long bones), spongy/cancellous bone (marrow cavities mein) — 2 main types. *[Q858]*
- 'Cartilage' joint par bone ki surface ko soft karta hai — Chondroblasts se produce hoti hai, nose-ear-trachea-larynx mein bhi milti hai. *[Q859]*
- Voluntary muscles 'hind limb' (legs) mein hoti hain — cylindrical fibers, bones-skin se attached, will (ichha) ke control mein. Examples: Diaphragm, pharynx, tongue, limb muscles. *[Q860]*
- 'Tendon' muscle ko bones se connect karta hai — collagen se bani fibrous connective tissue hai. Ligaments bone-to-bone connect karte hain, Cartilage joints-bones protect karti hai. *[Q861]*
- Muscle/muscular tissue body mein 'movements' ke liye responsible hai — Striated (voluntary/skeletal), Smooth (involuntary), Cardiac (branched, single nucleus, heart mein) — 3 types. *[Q862]*
- 'Bone' ek connective tissue hai — connective tissue animal ke 4 primary tissue types mein se ek hai (epithelial, muscle, nervous ke saath). Vascular bundles PLANTS mein milte hain (xylem-phloem). *[Q863]*
- 'Ligament' bones ko connect karta hai — bones-joints-organs ko jagah par hold karta hai. Areolar organs ko connect-surround karta hai, Tendon muscles ko bones se, Cartilage joints-bones ko protect karta hai. *[Q864]*
- Muscle tissue ke 3 types hote hain: (1) Cardiac — heart ki walls mein, striated, involuntary; (2) Smooth — hollow visceral organs mein (heart chhodkar), spindle-shaped, involuntary; (3) Skeletal — skeleton se attached, voluntary movements. *[Q865]*